

Dive With Data

Places of Worship Dashboard — Technical Documentation

1. Overview

This dashboard explores places of worship across France (Metropolitan France and the French Overseas Territories), covering their religious affiliation, denomination, and geographic distribution. It surfaces four key metrics and four visualizations, all driven by a single exported dataset and refined through cascading Country → Region → Department → City filters.

2. Data Source

The dashboard reads a single static file, `exported_data.json`, served from the app's public assets. It contains one record per place-of-worship location (35,084 records at the time of writing), each describing a location and the religious denominations recorded there.

3. Data Dictionary

Each record in `exported_data.json` has the following shape:

Top-level fields

Field	Type	Description
Country	string	Country name. Currently always "France".
Location_Type	string	Either "Metropolitan France" or "French Overseas Territories". Drives the dashboard's overseas/metropolitan toggle.
Details	object	Nested object with the geographic and religious details described below.

Details object

Field	Type	Description
Region	string	Administrative region (e.g. "Île-de-France", "Guadeloupe").
Department_Code	string	INSEE department code (e.g. "971").
Department_Name	string	Department name (e.g. "Guadeloupe").
Postcode	number	Postal code of the location.
City	string	City / commune name.
Population	number	Population of the city, from the underlying census source.
Surface	number	Surface area of the city, in hectares.
Inscription_Date_Count	number	Count of nationally-registered heritage inscriptions at this location. Summed for the "National Heritage Sites Registered" KPI.

Religious_Data	array	List of religion/denomination counts at this location (see below).
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Religious_Data entries

Field	Type	Description
Religion_Grouped	string	Broad religious grouping (e.g. "christian", "muslim", "jewish", "buddhist").
Denomination_Grouped	string	Denomination within that religion (e.g. "catholic", "protestant", "jehovahs witness").
Count	number	Number of places of worship of this denomination at this location.

4. Filters

Filters cascade in this order: Overseas/Metropolitan toggle → Country → Region → Department → City. Selecting a value at one level narrows the options shown at the levels below it; clearing a level resets everything beneath it. All four charts and all four KPI values respond to the same set of active filters.

5. KPI Definitions

- **Number of Places of Worship** — sum of every Count across all Religious_Data entries in the currently filtered set.
- **Average Place of Worship** — total places of worship divided by the number of locations in the filtered set.
- **National Heritage Sites Registered** — sum of Inscription_Date_Count across the filtered locations.
- **Diversity Index** — Shannon entropy computed over the religion mix of the filtered set (see Section 6).

6. Diversity Index Methodology

The diversity index is the Shannon entropy of the religion distribution within a group of locations (a region, department, or city, depending on the active filter level):

$$H = -\sum p_i \times \ln(p_i)$$

where p_i is the share of places of worship belonging to religion i , out of the total for that group. A value of 0 means every place of worship belongs to a single religion; higher values indicate a more even mix across religions. This is computed independently in two places in the codebase (the KPI panel and the Diversity Index chart) using the same formula, applied at the currently selected geographic granularity (Region, Department, or City).

7. Charts

- **Distribution by Religion/Denomination** (bubble/scatter) — the top 5 religions in the filtered set, plotted against every denomination present, sized by count.
- **Distribution of Places of Worship by Religion** (bar) — total count for the top 5 religions in the filtered set.

- **Religion Distribution** (heat map) — percentage share of each religion, broken down by Region, Department, or City depending on the active filter depth.
- **Diversity of Places of Worship by Religion** (bar) — the Shannon diversity index for each Region, Department, or City in the filtered set, sorted ascending.

For the underlying raw data, see the "Exported Data" download on the Resources page.